

ASSIGNMENT COVERSHEET

UTS: ENGINEERING & INFORMATION TECHNOLOGY		
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ASSESSMENT ITEM NUMBER & TITLE Assignment 2 - Release 1 Technical Report		
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Project Overview and Release 1 Scope

Release 0 delivered five containerised features, each with its own frontend, backend and database linked from the unified home page with a shared CSS theme. All services deploy from one docker-compose.yml. Each student has a .yml workflow, and every feature has an Ollama-backed AI-Mode.

Release 1 adds one shared MCP server (port 5201) and one shared RAG server (5200). This is run on the host outside Docker alongside Ollama and reached through host.docker.internal. Frontends only use them through their own backend.

Each student added their own feature updates incorporating the MCP and RAG servers in their frontend and backend, configured their docker-compose.yml service, and kept their workflow passing. Each student's specific updates are listed in the Features Updates/Release 1 Work table.

The group is responsible for the shared MCP and RAG servers, the agentic loop and the docker-compose.yml connection settings.

Release 1 Requirements

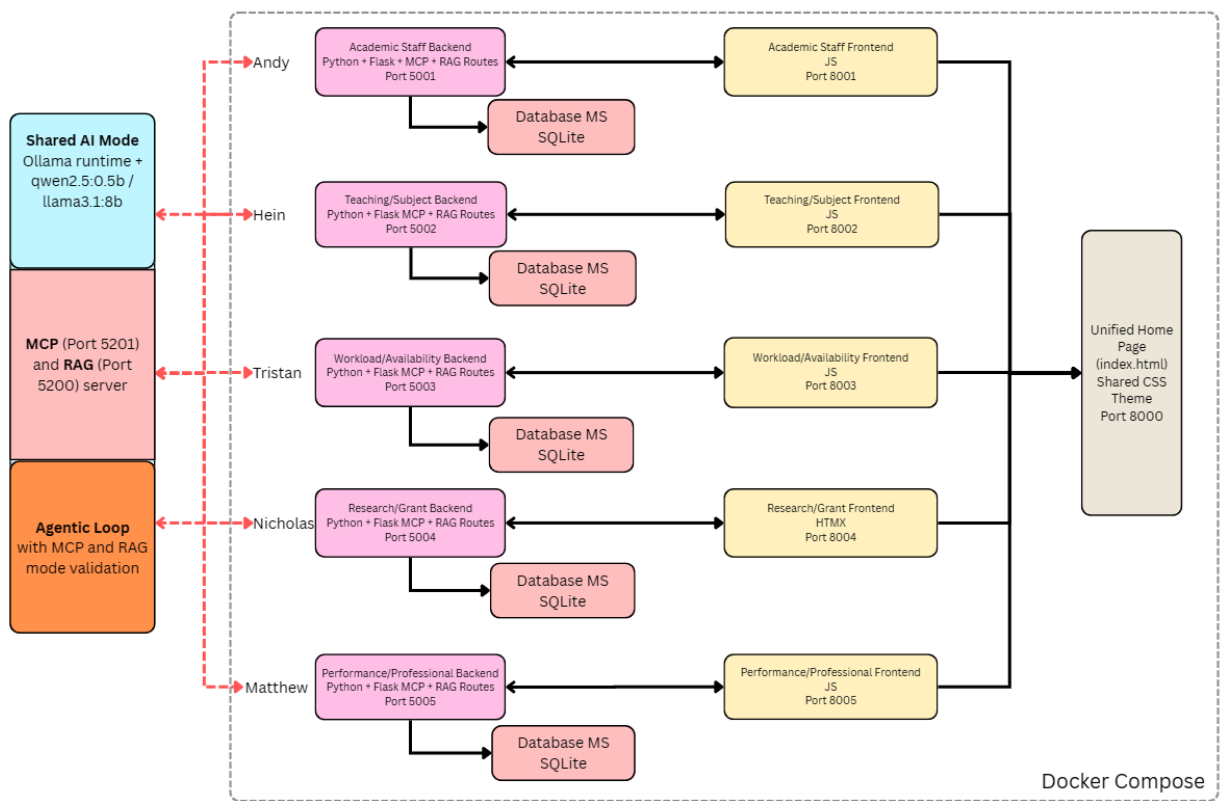
ID	Functional Requirement
FR1	The common local MCP server shall expose feature-owned tools via each feature's backend/API, with validated inputs and structured results/errors.
FR2	Feature frontends shall access MCP via their own backend, depending on whether MCP is enabled.
FR3	Shared local RAG server shall aggregate context provided by features and support corpus refresh, retrieval and grounded question answering.
FR4	RAG answers shall supply the users with citations and confidence level; insufficient evidence shall return an 'insufficient-context' response.
FR5	Feature frontends shall use their backend/API to access RAG and handle unavailable services gracefully.
FR6	The shared agentic loop shall validate MCP and RAG on real-time interactions and generate validation evidence about tools, retrieval, grounding and error handling.

Non-Functional Requirements

ID	Non-Functional Requirement	Justification
NFR1	Security: requests must have valid inputs, and responses must be displayed safely. Data and model processing should be local.	Reduces invalid requests and unsafe displayed content. Test incorrect input types, empty values and HTML text. Authentication is not implemented, so the system is intended for a trusted local environment.
NFR2	MCP tool boundaries: Tools must perform their defined read or validation operations through the relevant feature's backend/API.	Preserves service boundaries and prevents unintended changes. Check registered tool names, arguments and that read-only calls leave database records unchanged.
NFR3	RAG grounding: Answers must use retrieved context and return "Insufficient context" when supporting evidence is unavailable.	Reduces unsupported answers. Test questions with known evidence and unrelated questions, and report retrieval precision@5 and recall@5.
NFR4	Traceability: Supported RAG answers must include source citations and a confidence category.	Allows users to inspect the evidence behind an answer. Check source IDs, chunk metadata and shared audit records. Confidence describes retrieved evidence and does not guarantee correctness.
NFR5	Reliability: Shared-service failures must produce controlled errors while ordinary feature operations remain usable.	Keeps dependency failures understandable. Test unavailable services, invalid responses and disabled modes. Student 5 returns HTTP 503 when MCP/RAG is disabled or unavailable.
NFR6	Performance: External service requests must use configured timeouts to limit waiting.	Prevents indefinite requests. MCP downstream HTTP calls default to 10 seconds; Student 5 uses a 20-second MCP client timeout and a 120-second RAG answer timeout. Actual response times must be measured separately.
NFR7	Usability: Frontends must display readable results, loading states, errors, citations and confidence.	Helps users understand requests and results. Check successful MCP/RAG interactions and insufficient-context responses

		through every feature's UI. Disable submission buttons while requests are pending.
NFR8	Maintainability: Clients, tools and context loaders should be separated, with configurable service addresses and mode flags.	Makes changes easier to manage. Require passing automated tests and frontend/backend/database builds, with AI-Mode, MCP and RAG disabled during CI.

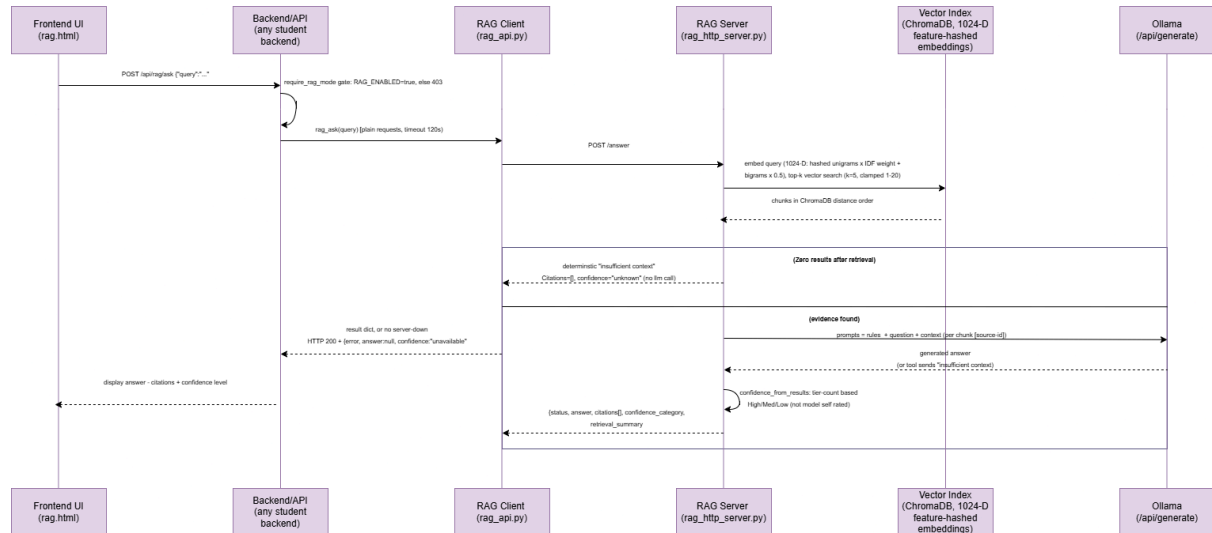
Release 1 Architecture



This architecture is extended from Release 0's architecture, in which the application is fully containerised within Docker Compose. Each feature runs its backend, frontend and SQLite database microservice with the frontend linking to the unified home page (port 8000), and sharing the group CSS theme. With Release 1, in addition to the shared AI service, the MCP (port 5201) and RAG (port 5200) server along with the agentic loop including MCP and RAG validation connects to the containerised application through each member's backend. This is shown through the red dashed arrows crossing the Docker Compose boundary reaching the host services of a member's backend through host.docker.internal. The connection is double-ended with backend calling MCP, RAG and Ollama, while MCP tools alongside RAG loaders call backend APIs for feature data.

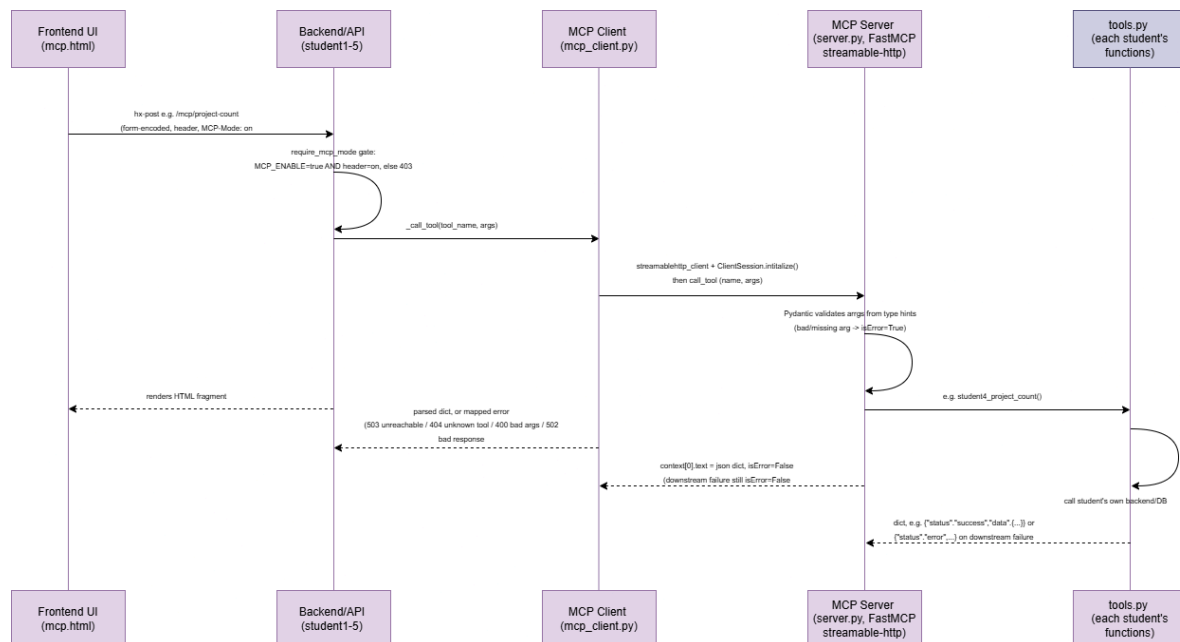
MCP and RAG Design

RAG Diagram



Explanation: The frontend will POST a question to the backend, checking the RAG_ENABLED gate and proceeding to send it to the RAG server. The server will embed the question, search the ChromaDB vector index for the closest relating chunks, then drop any that don't share any words / context with the question. If nothing is left, "Insufficient context" is returned with no citations and no call to the model. Otherwise, it will pass the remaining relevant chunks to the local LLM, responding to the question with citations and a confidence category based on the context provided and how much evidence was retrieved.

MCP Diagram



Explanation: Each student's featured frontend will send a request to their own backend. The backend checks the MCP_ENABLED gate, returning a 403 if it's off. If it's on, the backend's MCP client opens a streamable-http session that calls the named tool from the shared MCP server. The server will then validate the arguments so if a missing argument is rejected before reaching any code. The tool then calls the student's own backend, returning a json-like result. Moreover, if the downstream feature fails, the tool will return a structured error rather than crashing.

Validation and Results

Validation Approach

The validation approach for Release 1 involved testing each feature's MCP and RAG implementation through its frontend and corresponding backend/APIs. Additionally, the terminal output was used to validate these shared services along with the shared agentic loop. Lastly, each student's YAML workflow showed their features were built and passed with MCP and RAG disabled.

Results/Evidence

MCP server running locally on port 5201, listing all 14 tools contributed by each member.

```
(venv) C:\Users\kring\Documents\GitHub\faculty-management-system-ASD\ai-services\mcp-server>python server.py
Starting Faculty Management System MCP Server...
Server status: RUNNING
Available tools:
- student1_get_staff_profile
- student1_search_staff_by_expertise
- student2_validate_teaching_allocation
- student2_check_classroom_availability
- student3_staff_count
- student3_staff_by_status
- student3_staff_workload_detail
- student3_open_alerts
- student4_project_count
- student4_projects_by_department
- student4_project_grants_summary
- student4_research_history
- student5_staff_development_summary
- student5_training_by_skill_area
INFO:      Started server process [50596]
INFO:      Waiting for application startup.
[10/06/26 18:18:12] INFO      StreamableHTTP      streamable http manager.py:16
```

RAG server running locally on port 5200 outside docker. It serves /refresh, /retrieve and /answer for every feature.

```
(venv) C:\Users\kring\Documents\GitHub\faculty-management-system-ASD\ai-services\rag-server>python rag_http_server.py
RAG HTTP server running on 0.0.0.0:5200
```

List of all 16 containers running excluding Ollama running from release 0
docker-compose.yml. Ollama, MCP and RAG containers are not defined as they're reached
through host.docker.internal

NAME	IMAGE	COMMAND	SERVICE	CREATED
STATUS	PORTS			
faculty-management-system-asd-allocation-backend-1	faculty-management-system-asd-allocation-backend	"python app.py"	allocation-backend	3 minutes ag
o Up 3 minutes	0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp			
faculty-management-system-asd-allocation-database-1	faculty-management-system-asd-allocation-database	"python app.py"	allocation-database	3 minutes ag
o Up 3 minutes	6002/tcp			
faculty-management-system-asd-allocation-frontend-1	faculty-management-system-asd-allocation-frontend	"/docker-entrypoint..."	allocation-frontend	3 minutes ag
o Up 3 minutes	0.0.0.0:8002->80/tcp, [::]:8002->80/tcp			
faculty-management-system-asd-staff-backend-1	faculty-management-system-asd-staff-backend	"python app.py"	staff-backend	3 minutes ag
o Up 3 minutes	0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp			
faculty-management-system-asd-staff-database-1	faculty-management-system-asd-staff-database	"tail -f /dev/null"	staff-database	3 minutes ag
o Up 3 minutes				
faculty-management-system-asd-staff-frontend-1	faculty-management-system-asd-staff-frontend	"/docker-entrypoint..."	staff-frontend	3 minutes ag
o Up 3 minutes	0.0.0.0:8001->80/tcp, [::]:8001->80/tcp			
student3-backend	faculty-management-system-asd-student3-backend	"python backend-serv..."	student3-backend	3 minutes ag
o Up 3 minutes	0.0.0.0:5003->5003/tcp, [::]:5003->5003/tcp			
student3-database	faculty-management-system-asd-student3-database	"python app.py"	student3-database	3 minutes ag
o Up 3 minutes	0.0.0.0:5103->5103/tcp, [::]:5103->5103/tcp			
student3-frontend	faculty-management-system-asd-student3-frontend	"/docker-entrypoint..."	student3-frontend	3 minutes ag
o Up 3 minutes	0.0.0.0:8003->80/tcp, [::]:8003->80/tcp			
student4-backend	faculty-management-system-asd-student4-backend	"python app.py"	student4-backend	3 minutes ag
o Up 3 minutes	0.0.0.0:5004->5004/tcp, [::]:5004->5004/tcp			
student4-database	faculty-management-system-asd-student4-database	"python app.py"	student4-database	3 minutes ag
o Up 3 minutes	0.0.0.0:5104->5104/tcp, [::]:5104->5104/tcp			
student4-frontend	faculty-management-system-asd-student4-frontend	"python app.py"	student4-frontend	3 minutes ag
o Up 3 minutes	0.0.0.0:8004->8004/tcp, [::]:8004->8004/tcp			
student5-backend	faculty-management-system-asd-student5-backend	"python app.py"	student5-backend	3 minutes ag
o Up 3 minutes (healthy)	0.0.0.0:5005->5005/tcp, [::]:5005->5005/tcp			
student5-database	faculty-management-system-asd-student5-database	"python app.py"	student5-database	3 minutes ag
o Up 3 minutes (healthy)	0.0.0.0:5105->5105/tcp, [::]:5105->5105/tcp			
student5-frontend	faculty-management-system-asd-student5-frontend	"/docker-entrypoint..."	student5-frontend	3 minutes ag
o Up 3 minutes	0.0.0.0:8005->80/tcp, [::]:8005->80/tcp			
unified-home	nginx:alpine	"/docker-entrypoint..."	home	5 days ago
o Up 3 minutes	0.0.0.0:8000->80/tcp, [::]:8000->80/tcp			

Agentic Loop (MCP Validation Mode) - Report is stored in reports/mcp-validation-report.md

Select an option> 6

Running MCP

[MCP][OBSERVE] Collecting shared MCP evidence.
[MCP][ACT] Running implementation assessment.
[MCP][ADAPT] Running review model.

OBSERVE:

MCP VALIDATION EVIDENCE

- Structural files: PASS
- Transport: PASS (mcp.run(transport="streamable-http"))
- Statically registered tools: PASS (14 in server.py, contributing students: 1, 2, 3, 4, 5)
- requirements.txt declares mcp package: PASS
- Live tool listing: PASS (14 tool(s) reachable over real MCP protocol at http://localhost:5201/mcp)
- Live tool invocation (3 of 14 tools safely callable with no arguments; 11 require arguments and were not blindly invoked):

DETERMINISTIC VALIDATION:

Status: PASS

Functional validation: PASS (structure, transport, registered tools, requirements, live tool listing, live invocation, unknown-tool error handling)

Student coverage (registered): 1, 2, 3, 4, 5

Live tool invocation: 3/3 zero-argument tools invoked cleanly, 0 downstream feature-service failure(s), 0 MCP-transport failures

Student coverage (live-invoked): 3, 4

IMPLEMENTATION:

Status: PASS

Strengths: 14 tools are registered statically, contributing students: 1, 2, 3, 4, 5

Gaps: None observed

Review Scope:

Shared MCP Server Integration

```
REVIEW:
Risk: None observed
Correction: None
Retest: No additional retest required

[MCP][RECORD] Report written to C:\Users\kring\Documents\GitHub\faculty-management-system-ASD\reports\mcp-validation-report.md
=====
MCP validation run complete.
Report: C:\Users\kring\Documents\GitHub\faculty-management-system-ASD\reports\mcp-validation-report.md
=====
```

Agentic Loop (RAG Validation Mode) - Report is stored in reports/rag-validation-report.md

```
Select an option> 7

Running RAG

[RAG][OBSERVE] Collecting shared RAG evidence.
[RAG][ACT] Running implementation assessment.
[RAG][ADAPT] Running review model.

OBSERVE:
RAG VALIDATION EVIDENCE
- Structural files: PASS
- Required tools: PASS (refresh corpus, retrieve context, answer question)
- Live health: PASS ({'status': 'ok', 'service': 'faculty-management-rag'})
- Corpus refresh: PASS (207 chunks, vector store ready)
- Retrieval: PASS (5 chunk(s), mode=vector)
- Validation query: Which staff member has expertise in machine learning?
- Grounded answer: John Smith (staff ID 1) has expertise in machine learning.
- Citations: PASS (5 returned)
- Confidence: PASS (High)
- Insufficient-context behaviour: PASS
- Audit logging: PASS (6 new record(s))
- Retrieval metrics: 20 benchmark(s), mean P@5=0.470, mean R@5=0.975, minimum R@5=0.500

DETERMINISTIC VALIDATION:
Status: PASS
Functional validation: PASS (structure, tools, health, corpus refresh, retrieval, grounded answer, citations, confidence, insufficient-context handling, audit logging)
Retrieval quality: mean P@5=0.470, mean R@5=0.975, minimum R@5=0.500
Observed recall limitation: minimum R@5=0.500, so at least one benchmark did not retrieve all expected relevant evidence at k=5.

IMPLEMENTATION:
Status: PASS
Strengths: The RAG implementation is operational, grounded, measurable, and suitable for Release 1.

Gaps: None observed.

REVIEW:
Risk: None observed
Correction: None
Retest: No additional retest required
```

Student 1 (Andy Lam)

MCP Tools

Runs a registered tool on the shared MCP server and shows its structured result.

GET STAFF PROFILE – STAFF ID

Run tool

SEARCH STAFF BY EXPERTISE

Run tool

Tool: student1_get_staff_profile

SUCCESS

Marcus Chen

Senior Lecturer — Information Technology — Full-time, Active

Expertise

- Software Architecture (skill level 5/5)

Qualifications

- PhD in Software Engineering, UNSW (2012)

Availability

- Wednesday 13:00-15:00 — Available

▼ Raw tool result (JSON)

```
{
  "data": {
    "availability": [
      {
        "availability_id": 3,
        "availability_status": "Available",
        "day": "Wednesday",
        "time_slot": "13:00-15:00"
      }
    ],
    "expertise": [
      {
        "expertise_area": "Software Architecture",
        "expertise_id": 3
      }
    ]
  }
}
```

```
(venv) C:\Users\kring\Documents\GitHub\faculty-management-system-ASD>curl.exe -X POST localhost:5001/api/mcp/staff-profile -H "Content-Type: application/json" -d '{"staff_id": 3}'
```

```
{
  "data": {
    "availability": [
      {
        "availability_id": 3,
        "availability_status": "Available",
        "day": "Wednesday",
        "time_slot": "13:00-15:00"
      }
    ],
    "expertise": [
      {
        "expertise_area": "Software Architecture",
        "expertise_id": 3,
        "skill_level": 5
      }
    ],
    "qualifications": [
      {
        "institution": "UNSW",
        "qualification_id": 3,
        "qualification_name": "PhD in Software Engineering",
        "year_obtained": 2012
      }
    ],
    "staff": {
      "department_id": 2,
      "department_name": "Information Technology",
      "email": "marcus.chen@university.edu",
      "employment_type": "Full-time",
      "name": "Marcus Chen",
      "phone": "0412345680",
      "position": "Senior Lecturer",
      "staff_id": 3,
      "status": "Active"
    }
  },
  "status": "success",
  "tool": "student1_get_staff_profile"
}
```

Ask About Staff (RAG)

Answers only from retrieved staff records, with sources and a confidence rating.

QUESTION

Who has expertise in robotics

Ask

Refresh knowledge base

CONFIDENCE: HIGH

David Kim has expertise in robotics.

Sources

1. student1/staff/5 (staff_management, tier_1)
2. student2/subjects/41113 (teaching_subject_and_classroom_allocation, tier_1)
3. student2/subjects/41039 (teaching_subject_and_classroom_allocation, tier_1)
4. student2/subjects/31268 (teaching_subject_and_classroom_allocation, tier_1)
5. student2/subjects/41095 (teaching_subject_and_classroom_allocation, tier_1)
6. student2/subjects/41082 (teaching_subject_and_classroom_allocation, tier_1)
7. student2/subjects/31271 (teaching_subject_and_classroom_allocation, tier_1)
8. student2/subjects/41092 (teaching_subject_and_classroom_allocation, tier_1)

Ask About Staff (RAG)

Answers only from retrieved staff records, with sources and a confidence rating.

QUESTION

whats the capital of france

Ask

Refresh knowledge base

CONFIDENCE: INSUFFICIENT

Insufficient context.

No staff records relevant to this question were found, so no answer was generated.

← Student 1 CI

✓ Merge remote-tracking branch 'origin/main' into student-1-andy-staff... #95

Re-run all jobs ...

Summary

All jobs

build-and-validate

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

build-and-validate

succeeded 1 hour ago in 55s

Search logs

- Set up job 1s
- Run actions/checkout@v4 8s
- Set up Python 2s
- Install dependencies 22s
- Check backend imports correctly 1s
- Run unit tests (MCP/RAG servers failed or disabled) 1s
- Validate docker-compose MCP/RAG configuration 8s
- Build database Docker image 8s
- Build backend Docker image 11s
- Build frontend Docker image 2s
- Run containers and smoke test (MCP/RAG disabled) 2s
 - Container logs on failure 8s
- Post Set up Python 8s
- Post Run actions/checkout@v4 8s
- Complete job 8s

Student 2 (Hein Htet Aung)

Check Classroom Availability

Checks whether a classroom is free for a specified date and time without creating or modifying allocation data.

CLASSROOM ID

CB11.04.406

DATE

03/09

YEAR

2026

START TIME

02:00 PM

END TIME

04:30 PM

Check Availability

MCP TOOL

UNAVAILABLE

Classroom Availability

CLASSROOM

CB11.04.406

DATE

03/09/2026

TIME

14:00 - 16:30

TOOL

student2_check_classroom_availability

Conflicts with teaching allocation #4.

▼ View raw JSON

```
{
  "data": {
    "available": false,
    "conflicting_allocation_id": 4
  },
  "status": "success",
  "title": "Classroom Availability: CB11.04.406",
  "tool": "student2_check_classroom_availability"
}
```

```
(.venv) hein@Heins-MacBook-Pro faculty-management-system-ASD % curl -s -X POST http://localhost:5002/mcp /check-classroom-availability \
-H "X-MCP-Mode: on" \
-F "classroom_id=CB11.04.406" \
-F "date=03/09" \
-F "year=2026" \
-F "start_time=14:00" \
-F "end_time=16:30" | jq
{
  "data": {
    "available": false,
    "conflicting_allocation_id": 4
  },
  "status": "success",
  "title": "Classroom Availability: CB11.04.406",
  "tool": "student2_check_classroom_availability"
}
```

Validate Teaching Allocation

Validates a proposed teaching allocation using the existing subject, classroom, scheduling and staff rules. No teaching allocation is created or modified.

SUBJECT OFFER

4TH4_SPR_2026

CLASSROOM ID

CB11.04.406

DAY

Thursday

DATE RANGE

03/09 - 30/09

START TIME

11:00 AM

END TIME

04:30 PM

CLASS TYPE

CNR

EXPECTED CLASS SIZE

100

ASSIGNED STAFF

Ben Robertson

Ben Robertson · Senior Lecturer · Marketing

Clear selection

Optional. Leave unassigned if no staff member should be validated.

Validate Allocation

MCP TOOL

NEEDS ATTENTION

Teaching Allocation Validation

SUBJECT OFFER

4TH4_SPR_2026

CLASSROOM

CB11.04.406

SCHEDULE

THU 11:00 - 16:30

CLASS TYPE / SIZE

CNR · 100

Validation Issues

- Classroom is unavailable during one or more requested dates.
- Assigned staff member is unavailable during the requested time.

▼ View raw JSON

```
{
  "data": {
    "errors": [
      "Classroom is unavailable during one or more requested dates.",
      "Assigned staff member is unavailable during the requested time."
    ],
    "valid": false
  },
  "status": "success",
  "title": "Teaching Allocation Validation",
  "tool": "student2_validate_teaching_allocation"
}
```

```
• (.venv) hein@Heins-MacBook-Pro faculty-management-system-ASD % curl -s -X POST http://localhost:5002/mcp/validate-teaching-allocation \
-H "X-MCP-Mode: on" \
-F "offer_id=41114_SPR_2026" \
-F "classroom_id=CB11.04.406" \
-F "day=THU" \
-F "date_range=03/09 - 30/09" \
-F "start_time=11:00" \
-F "end_time=16:30" \
-F "class_type=CNR" \
-F "expected_class_size=100" \
-F "assigned_staff_member=7" | jq
{
  "data": {
    "errors": [
      "Classroom is unavailable during one or more requested dates.",
      "Assigned staff member is unavailable during the requested time."
    ],
    "valid": false
  },
  "status": "success",
  "title": "Teaching Allocation Validation",
  "tool": "student2_validate_teaching_allocation"
}
```

Ask Grounded Question

Retrieves relevant evidence and sends only that context to the local language model. The answer includes its retrieved citations and confidence category.

QUESTION

When can't I book CB11.04.406?

NUMBER OF CHUNKS

5

Answer with Citations

Answer

You can't book CB11.04.406 on THU during 03/08 - 20/09 2026, from 14:00 to 16:00, as it is already allocated for Advanced Software Development (subject 41114) during that scheduled time.

Confidence: High

Citations

- student2/classrooms/CB11.04.406 (tier_1)
- student2/classrooms/CB11.04.406/allocations (tier_1)
- student4/publications/9 (tier_1)
- student2/teaching-allocations/4 (tier_1)

Ask Grounded Question

Retrieves relevant evidence and sends only that context to the local language model. The answer includes its retrieved citations and confidence category.

QUESTION

Which one of the staffs has cancer?

NUMBER OF CHUNKS

5

Answer with Citations

Answer

Insufficient context.

Confidence: High

Citations

- student5/development-goals/3 (tier_1)
- student5/development-goals/7 (tier_1)
- student2/classrooms/CB10.02.302/allocations (tier_1)
- student2/subjects/31271/allocations (tier_1)

← Student 2 - Teaching, Subject and Classroom Allocation

Student 2 - Teaching, Subject and Classroom Allocation #177

Re-run all jobs

...

Summary

All jobs

Unit and code validation

Build Docker images

Runtime and API smoke tests

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Unit and code validation

succeeded 1 minute ago in 22s

Search logs

Refresh

Settings

Set up job

0s

Check out repository

1s

Set up Python 3.11

2s

Install Student 2 dependencies

11s

Validate Python syntax

0s

Validate backend import

2s

Verify AI prompt files

0s

Run Student 2 automated tests

1s

Post Set up Python 3.11

0s

Post Check out repository

1s

Complete job

0s

← Student 2 - Teaching, Subject and Classroom Allocation

Student 2 - Teaching, Subject and Classroom Allocation #177

Re-run all jobs

...

Summary

All jobs

Unit and code validation

Build Docker images

Runtime and API smoke tests

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Build Docker images

succeeded 1 minute ago in 29s

Search logs

Refresh

Settings

Set up job

0s

Check out repository

1s

Build database Docker image

11s

Build backend Docker image

11s

Build frontend Docker image

3s

Confirm Docker images exist

0s

Post Check out repository

0s

Complete job

0s

Summary

All jobs

Unit and code validation

Build Docker images

Runtime and API smoke tests

Run details

Usage

Workflow file

1 warning and 1 notice

Runtime and API smoke tests

succeeded 1 minute ago in 40s

Search logs

🔄 ⚙️

> ⌚ Set up job1s

> ⌚ Check out repository1s

> ⌚ Build Student 2 Docker images27s

> ⌚ Create isolated CI network0s

> ⌚ Start database service0s

> ⌚ Wait for database service2s

> ⌚ Start backend service0s

> ⌚ Wait for backend service2s

> ⌚ Start frontend service0s

> ⌚ Wait for frontend service2s

> ⌚ Verify seeded database records0s

> ⌚ Verify backend resource endpoints0s

> ⌚ Verify Subject CRUD lifecycle0s

> ⌚ Verify invalid Subject validation1s

> ⌚ Verify classroom capacity validation0s

> ⌚ Verify back-to-back scheduling rule0s

> ⌚ Verify MCP and RAG are disabled during CI/CD0s

> ⌚ Verify frontend files are served0s

> ⌚ Collect container logs0s

> ⌚ Create CI evidence summary0s

> ⌚ Upload CI evidence1s

> ⌚ Clean up CI containers0s

> ⌚ Post Check out repository0s

> ⌚ Complete job0s

Student 3 (Tristan Lim)

Staff Count

STATUS (OPTIONAL)

DEPARTMENT (OPTIONAL)

overloaded

Computer Science

Get Staff Count

Staff Count

FIELD	VALUE
Count	3

```
(venv-run) PS D:\Coding Projects\faculty-management-system-450\student-3\Tristan-Lim\backend-service & "D:\Coding Projects\faculty-management-system-450\venv-run\Scripts\python.exe" -c "from services.mcp_client import mcp_invoke; import json; print(json.dumps(mcp_invoke('student3_staff_count', {}), indent=2))"
{
  "status": "success",
  "data": {
    "count": 10
  }
}
```

Staff by Status

STATUS

Overloaded

Get Staff by Status

Staff with status overloaded

CONTRACTED FRACTION	CURRENT TOTAL HOURS	DEPARTMENT	MAX WEEKLY HOURS	PROFILE ID	SEMESTER	STAFF ID	STAFF NAME	STATUS
1.0	41.0	Computer Science	37.5	1	2026-S2	1	John Smith	overloaded
0.5	20.0	Information Technology	18.75	4	2026-S2	4	Priya Nair	overloaded
0.5	20.0	Management	18.75	8	2026-S2	8	Hana Suzuki	overloaded

```
(.venv-run) PS D:\Coding Projects\faculty-management-system-ASD\student-3\Tristan-Lin\backend-service> & "D:\Coding Projects\faculty-management-system-ASD\venv-run\Scripts\python.exe" -c "from services.mcp_client import mcp_invoke; import json; print(json.dumps(mcp_invoke('student3_staff_by_status', {'status': 'overloaded'}), indent=2))"
{
  "status": "success",
  "http_status": 200,
  "url": "http://localhost:5003/api/staff-workload/profiles",
  "data": [
    {
      "contracted_fraction": 1.0,
      "current_total_hours": 41.0,
      "department": "Computer Science",
      "max_weekly_hours": 37.5,
      "profile_id": 1,
      "semester": "2026-S2",
      "staff_id": 1,
      "staff_name": "John Smith",
      "status": "overloaded"
    },
    {
      "contracted_fraction": 0.5,
      "current_total_hours": 20.0,
      "department": "Information Technology",
      "max_weekly_hours": 18.75,
      "profile_id": 4,
      "semester": "2026-S2",
      "staff_id": 4,
      "staff_name": "Priya Nair",
      "status": "overloaded"
    },
    {
      "contracted_fraction": 0.5,
      "current_total_hours": 20.0,
      "department": "Management",
      "max_weekly_hours": 18.75,
      "profile_id": 8,
      "semester": "2026-S2",
      "staff_id": 8,
      "staff_name": "Hana Suzuki",
      "status": "overloaded"
    }
  ]
}
```

Staff Workload Detail

STAFF ID

1

Get Workload Detail

Workload Detail for Staff #1

FIELD	VALUE
Contracted Fraction	1.0
Current Total Hours	41.0
Department	Computer Science
Max Weekly Hours	37.5
Profile Id	1
Semester	2026-S2
Staff Id	1
Staff Name	John Smith
Status	overloaded

```
(.venv-run) PS D:\Coding Projects\faculty-management-system-ASD\student-3\Tristan-Lin\backend-service> & "D:\Coding Projects\faculty-management-system-ASD\venv-run\Scripts\python.exe" -c "from services.mcp_client import mcp_invoke; import json; print(json.dumps(mcp_invoke('student3_staff_workload_detail', {'staff_id': 1}), indent=2))"
{
  "status": "success",
  "data": {
    "contracted_fraction": 1.0,
    "current_total_hours": 41.0,
    "department": "Computer Science",
    "max_weekly_hours": 37.5,
    "profile_id": 1,
    "semester": "2026-S2",
    "staff_id": 1,
    "staff_name": "John Smith",
    "status": "overloaded"
  }
}
```

Open Alerts

DEPARTMENT (OPTIONAL)

Computer Science

Get Open Alerts

Open Alerts in Computer Science

ALERT ID	ALERT TYPE	DATE RAISED	MESSAGE	SEVERITY	STAFF ID	STATUS
1	overload	2026-08-20	Total 41.0h exceeds cap 37.5h for Computer Science	high	1	open
2	clash	2026-08-20	Conference leave overlaps 31266 teaching weeks	medium	1	open

```
(.venv-run) PS D:\Coding Projects\faculty-management-system-ASD\student-3\Tristan-Lin\backend-service> & "D:\Coding Projects\faculty-management-system-ASD\venv-run\Scripts\python.exe" -c "from services.mcp_client import mcp_invoke; import json; print(json.dumps(mcp_invoke('student3_open_alerts', {'department': 'Computer Science'}), indent=2))"
{
  "status": "success",
  "data": [
    {
      "alert_id": 1,
      "alert_type": "overload",
      "date_raised": "2026-08-20",
      "message": "Total 41.0h exceeds cap 37.5h for Computer Science",
      "severity": "high",
      "staff_id": 1,
      "status": "open"
    },
    {
      "alert_id": 2,
      "alert_type": "clash",
      "date_raised": "2026-08-20",
      "message": "Conference leave overlaps 31266 teaching weeks",
      "severity": "medium",
      "staff_id": 1,
      "status": "open"
    }
  ]
}
```

ask

QUESTION

Which staff members are overloaded?

Answer with Citations

The staff members who are overloaded are: 1. John Smith (staff ID 1) 2. Hana Suzuki (staff ID 8) 3. Priya Nair (staff ID 4)

Confidence: HIGH

CHUNK	SOURCE	TIER	FEATURE	STUDENT
student3_profile_1	student3/profiles/1	tier_1	workload_and_availability_management	3
student3_profile_8	student3/profiles/8	tier_1	workload_and_availability_management	3
student3_profile_4	student3/profiles/4	tier_1	workload_and_availability_management	3
student3_alert_9	student3/alerts/9	tier_1	workload_and_availability_management	3
student3_alert_8	student3/alerts/8	tier_1	workload_and_availability_management	3

```
PS D:\Coding Projects\faculty-management-system-ASD> curl.exe -X POST localhost:5003/rag/ask -H "Content-Type: application/json" -d '{"query": "Which staff members are overloaded?", "k": 5}'
{
  "answer": "John Smith, Hana Suzuki, Priya Nair.",
  "citations": [
    {
      "authority_tier": "tier_1",
      "chunk_id": "student3_profile_1",
      "feature": "workload_and_availability_management",
      "source_id": "student3/profiles/1",
      "student": "3"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student3_profile_8",
      "feature": "workload_and_availability_management",
      "source_id": "student3/profiles/8",
      "student": "3"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student3_profile_4",
      "feature": "workload_and_availability_management",
      "source_id": "student3/profiles/4",
      "student": "3"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student3_alert_9",
      "feature": "workload_and_availability_management",
      "source_id": "student3/alerts/9",
      "student": "3"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student3_alert_8",
      "feature": "workload_and_availability_management",
      "source_id": "student3/alerts/8",
      "student": "3"
    }
  ],
  "confidence_category": "High",
  "query": "Which staff members are overloaded?",
  "retrieval_summary": {
    "k": 5,
    "retrieved_count": 5,
    "top_chunk": "student3_profile_1"
  },
  "status": "success"
}
```

ask

QUESTION

what are the ingredients inside a burrito?

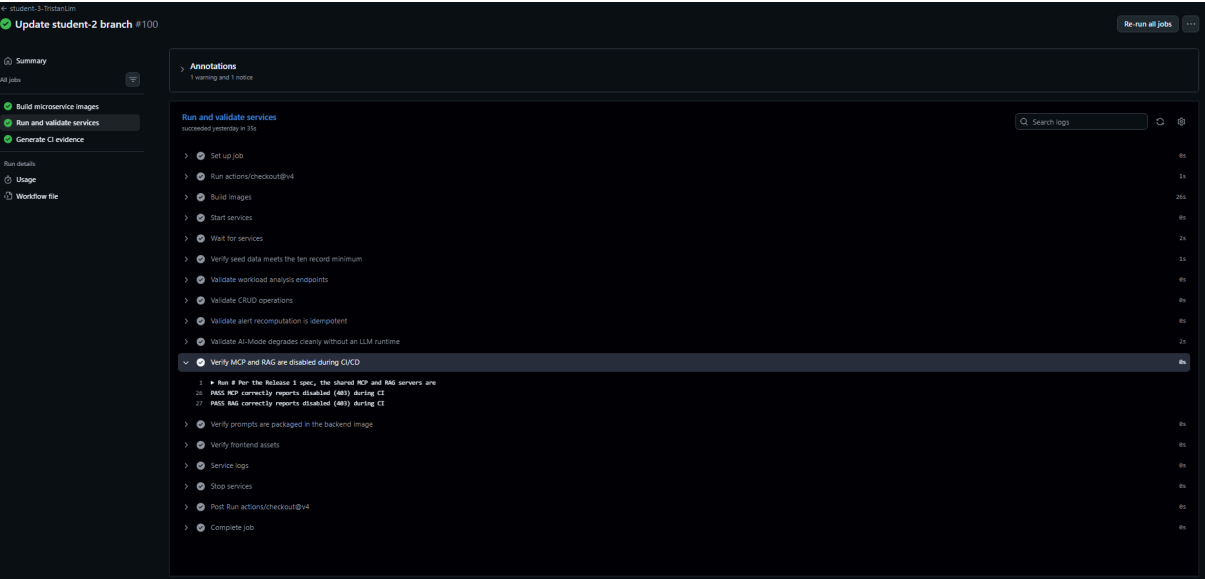
Answer with Citations

Insufficient context.

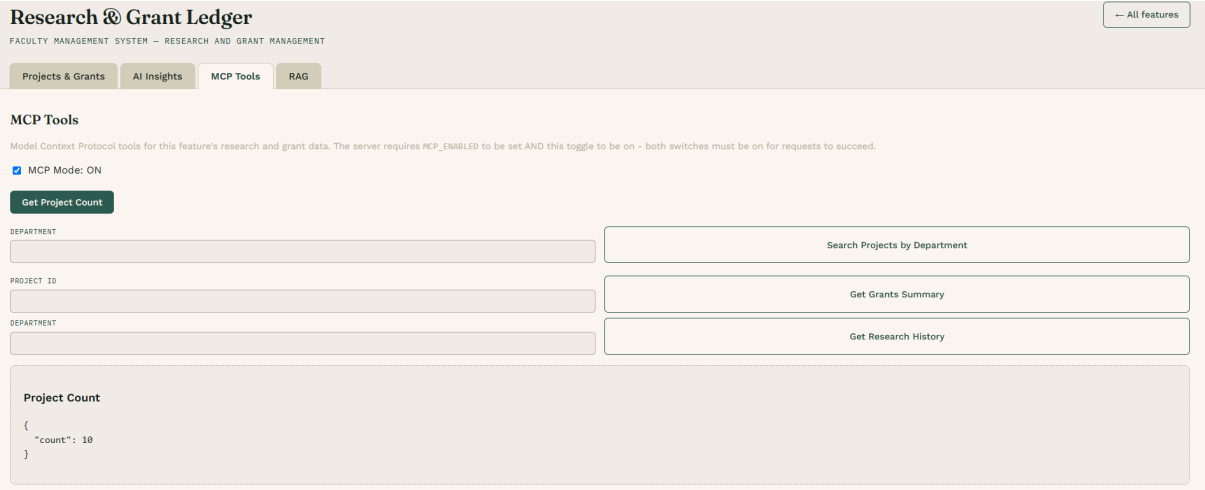
Confidence: UNKNOWN

No citations - Insufficient context.

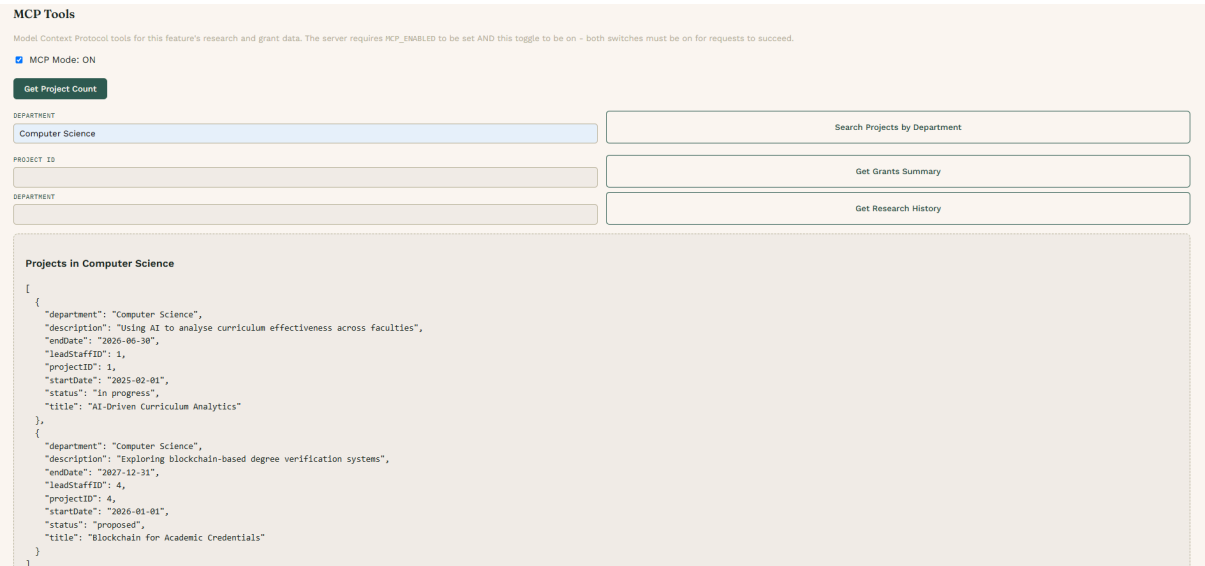
```
* (.venv-run) PS D:\Coding Projects\faculty-management-system-ASD> curl.exe -X POST localhost:5003/rag/ask -H "Content-Type: application/json" -d '{"query": "what are the ingredients inside a burrito?", "k": 5}'
{
  "answer": "Insufficient context.",
  "citations": [],
  "confidence_category": "Unknown",
  "query": "what are the ingredients inside a burrito?",
  "retrieval_summary": {
    "k": 5,
    "retrieved_count": 0,
    "top_chunk": null
  },
  "status": "success"
}
```



Student 4 (Nicholas Hatzidimitriou)



```
nicho@DESKTOP-LJT90TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$ curl -s -X POST http://localhost:5004/mcp/project-count -H "X-MCP-Mode: on"
<div class="mcp-result"><h3>Project Count</h3><pre>{
  "count": 10
}</pre></div>nicho@DESKTOP-LJT90TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$
```



```
nicho@DESKTOP-LJ190TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$ curl -s -X POST http://localhost:5004/mcp/projects-by-department -H "X-MCP-Mode: on" -d "department=Computer Science"
<div class="mcp-result"><h3>Projects in Computer Science</h3><pre>[
  {
    "department": "Computer Science",
    "description": "Using AI to analyse curriculum effectiveness across faculties",
    "endDate": "2026-06-30",
    "leadStaffID": 1,
    "projectID": 1,
    "startDate": "2025-02-01",
    "status": "in progress",
    "title": "AI-Driven Curriculum Analytics"
  },
  {
    "department": "Computer Science",
    "description": "Exploring blockchain-based degree verification systems",
    "endDate": "2027-12-31",
    "leadStaffID": 4,
    "projectID": 4,
    "startDate": "2026-01-01",
    "status": "proposed",
    "title": "Blockchain for Academic Credentials"
  }
]
```

Ask the Research Assistant

Ask a natural-language question about research projects, grants, or publications. The answer is grounded in retrieved evidence from the real corpus, with citations and a confidence rating shown alongside it - not just a bare answer.

Refresh Corpus

QUESTION

what staff is assigned to Financial Risk Modelling with AI, what is it about and what are the grants and publications attached to it?

Ask

The staff assigned to Financial Risk Modelling with AI is Hana Suzuki (staff ID 8). This project involves using AI-based approaches to predict financial market risk. Grants attached to this project include: * A grant from Fintech Research Partnership for \$110,000. Publications attached to this project include: * Predicting Market Volatility with Deep Learning (Journal of Financial Engineering, 2024-09-12) * Risk-Adjusted Portfolio Optimisation using AI (Quantitative Finance Review, 2024-06-22)

Confidence: High
Citations: student4/publications/6, student4/grants/9, student4/publications/10, student4/projects/9, student4/publications/4

Ask the Research Assistant

Ask a natural-language question about research projects, grants, or publications. The answer is grounded in retrieved evidence from the real corpus, with citations and a confidence rating shown alongside it - not just a bare answer.

Refresh Corpus

QUESTION

what is the capital of australia?

Ask

Insufficient context.

Confidence: Unknown
Citations: none

```
nicho@DESKTOP-LJ190TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$ curl -s -X POST http://localhost:5004/api/rag/ask \
-H "Content-Type: application/json" \
-d '{"query":"What is the Financial Risk Modelling with AI project about?"}' | python3 -m json.tool
{
  "answer": "The Financial Risk Modelling with AI project is about using AI-based approaches to predict financial market risk, as described in the project's description. This project was completed and led by staff member with ID 8, who also co-authored two publications related to this project: \"Predicting Market Volatility with Deep Learning\" and \"Risk-Adjusted Portfolio Optimisation using AI\"",
  "citations": [
    {
      "authority_tier": "tier_1",
      "chunk_id": "student4_grant_9",
      "feature": "research_grant_management",
      "source_id": "student4/grants/9",
      "student": "4"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student4_publication_6",
      "feature": "research_grant_management",
      "source_id": "student4/publications/6",
      "student": "4"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student4_publication_10",
      "feature": "research_grant_management",
      "source_id": "student4/publications/10",
      "student": "4"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student4_project_9",
      "feature": "research_grant_management",
      "source_id": "student4/projects/9",
      "student": "4"
    },
    {
      "authority_tier": "tier_1",
      "chunk_id": "student4_publication_4",
      "feature": "research_grant_management",
      "source_id": "student4/publications/4",
      "student": "4"
    }
  ],
  "confidence_category": "High",
  "query": "What is the Financial Risk Modelling with AI project about?",
  "retrieval_summary": {
    "k": 5,
    "retrieved_count": 5,
    "top_chunk": "student4_grant_9"
  },
  "status": "success"
}
```

```
(venv) nicho@DESKTOP-LJ190TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$ docker compose -f docker-compose.student4.yml up -d
[+] Running 3/3
 ✓ Container student4-database Running
 ✓ Container student4-backend Running
 ✓ Container student4-frontend Running
```

```
(venv) nicho@DESKTOP-LJ190TN:~/faculty-management-system-ASD/student-4Nicholas-Hatzidimitriou$ docker ps --filter name=student4 --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
NAME                STATUS                PORTS
student4-frontend    Up 51 minutes         0.0.0.0:8004->8004/tcp, [::]:8004->8004/tcp
student4-backend     Up 51 minutes         0.0.0.0:5004->5004/tcp, [::]:5004->5004/tcp
student4-database    Up 51 minutes         0.0.0.0:5104->5104/tcp, [::]:5104->5104/tcp
```

Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-... #120

Re-run all jobs

Summary

All jobs

Validate Python syntax

Validate database initialises and seeds correctly

Validate shared MCP/RAG wiring in docker-compose.yml

Build Docker images

Runtime smoke test (MCP/RAG disabled)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Validate Python syntax

succeeded 20 hours ago in 16s

Search logs

🔄 ⚙️

> Set up job

> Checkout repository

> Set up Python

> Install backend dependencies

> Compile-check backend

> Install database-service dependencies

> Compile-check database-service

> Install frontend dependencies

> Compile-check frontend

> Post Set up Python

> Post Checkout repository

> Complete job

Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-... #120

Re-run all jobs

Summary

All jobs

Validate Python syntax

Validate database initialises and seeds correctly

Validate shared MCP/RAG wiring in docker-compose.yml

Build Docker images

Runtime smoke test (MCP/RAG disabled)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Validate database initialises and seeds correctly

succeeded 20 hours ago in 7s

Search logs

🔄 ⚙️

> Set up job

> Checkout repository

> Set up Python

> Install database-service dependencies

> Run init_db.py and seed.py

> Verify seeded record counts

> Post Set up Python

> Post Checkout repository

> Complete job

Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-... #120

Re-run all jobs

Summary

All jobs

Validate Python syntax

Validate database initialises and seeds correctly

Validate shared MCP/RAG wiring in docker-compose.yml

Build Docker images

Runtime smoke test (MCP/RAG disabled)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Validate shared MCP/RAG wiring in docker-compose.yml

succeeded 20 hours ago in 6s

Search logs

🔄 ⚙️

> Set up job

> Checkout repository

> Validate student4-backend points at the shared host MCP/RAG servers

> Post Checkout repository

> Complete job

Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-... #120

Re-run all jobs

Summary

All jobs

Validate Python syntax

Validate database initialises and seeds correctly

Validate shared MCP/RAG wiring in docker-compose.yml

Build Docker images

Runtime smoke test (MCP/RAG disabled)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Build Docker images

succeeded 20 hours ago in 29s

Search logs

🔄 ⚙️

> Set up job

> Checkout repository

> Build backend image

> Build frontend image

> Build database-service image

> Post Checkout repository

> Complete job

✔ Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-... #120

Re-run all jobs

Summary

All jobs

✔ Validate Python syntax

✔ Validate database initialises and seeds correctly

✔ Validate shared MCP/RAG wiring in docker-compose.yml

✔ Build Docker images

✔ Runtime smoke test (MCP/RAG disabled)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

Runtime smoke test (MCP/RAG disabled)

succeeded 20 hours ago in 36s

Search logs

🔄 ⚙️

> ✔ Set up job1s

> ✔ Checkout repository2s

> ✔ Build images25s

> ✔ Create isolated CI network0s

> ✔ Start database service0s

> ✔ Start backend service (MCP/RAG disabled)0s

> ✔ Wait for backend health check2s

> ✔ Verify seeded project data is served0s

> ✔ Verify MCP is retained but cleanly disabled0s

> ✔ Verify RAG is retained but cleanly disabled0s

> ⚙️ Collect container logs on failure0s

> ✔ Clean up CI containers1s

> ✔ Post Checkout repository0s

> ✔ Complete job0s

Student 5 (Matthew Barnard)

PD

FACULTY MANAGEMENT SYSTEM

Performance & Professional Development

All features

Release 1

Reviews10

Active goals9

Training programs10

Pending recommendations9

Goals

Reviews

Training catalogue

Staff training

Recommendations

AI-Mode

MCP Tools

RAG Questions

SHARED MCP SERVER

MCP enabled

Development tools

Look up recorded development activity or find training programs by skill area.

Staff development summary

Staff ID

1

Get summary

Find training by skill area

Skill area

Leadership

Find programs

```
d -Method Post `
>> -Uri "http://localhost:5005/api/mcp/training-by-skill" `
>> -ContentType "application/json" `
>> -Body '{"skillArea":"Leadership"}' |
>> ConvertTo-Json -Depth 10
{
  "data": [
    {
      "description": "Leadership fundamentals for academic coordinators.",
      "endDate": "2026-09-24",
      "provider": "UTS Learning Hub",
      "skillArea": "Leadership",
      "startDate": "2026-09-10",
      "title": "Academic Leadership Foundations",
      "trainingID": 1
    }
  ],
  "match_count": 1,
  "query": "Leadership",
  "status": "success",
  "tool": "student5_training_by_skill_area"
}
```

PD

FACULTY MANAGEMENT SYSTEM

Performance & Professional Development

← All features

Release 1

Reviews

10

Active goals

9

Training programs

10

Pending recommendations

9

Goals

Reviews

Training catalogue

Staff training

Recommendations

AI-Mode

MCP Tools

RAG Questions

SHARED RAG SERVER

RAG enabled

Ask about performance and development

Ask a question about recorded reviews, goals, training or recommendations. Answers show their sources and confidence.

Your question

What is the progress and target date of the Strengthen academic leadership development goal?

Ask question

Refresh context

Retrieving context and generating an answer...

Matthew - Performance and Professional Development

student-5-MatthewBarnard.yml

Filter workflow runs

...

137 workflow runs				Event	Status	Branch	Actor
This workflow has a workflow_dispatch event trigger.				Run workflow			
✓ Merge pull request #61 from Hein-Htet-Aung08/student-5-matthew-perfor...	main	2 minutes ago	2m 11s	...			
Matthew - Performance and Professional Development #140: Commit 70b090e pushed by MattBarnard4							
✓ Use host Ollama in configuration	student-5-matthew-performan...	2 minutes ago	1m 25s	...			
Matthew - Performance and Professional Development #139: Pull request #61 opened by MattBarnard4							
✓ Use host Ollama in Student 5 configuration	student-5-matthew-performan...	22 minutes ago	1m 32s	...			
Matthew - Performance and Professional Development #138: Commit 7ab7295 pushed by MattBarnard4							
✓ Merge remote-tracking branch 'origin/main' into student-1-andy-staff-...	student-1-andy-staff-manage...	Oct 6, 21:10 GMT+11	1m 25s	...			
Matthew - Performance and Professional Development #137: Commit 9060c06 pushed by Alamb-beyondor							
✓ Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-...	main	Oct 6, 18:54 GMT+11	1m 15s	...			
Matthew - Performance and Professional Development #136: Commit 894108b pushed by Hein-Htet-Aung08							
✓ Student 2 hein teaching subject and classroom allocation	student-2-hein-teaching-sub...	Oct 6, 18:54 GMT+11	1m 37s	...			
Matthew - Performance and Professional Development #135: Pull request #58 opened by Hein-Htet-Aung08							
✓ Merge remote-tracking branch 'origin/main' into student-2-hein-teachi...	student-2-hein-teaching-sub...	Oct 6, 18:53 GMT+11	1m 18s	...			
Matthew - Performance and Professional Development #134: Commit 9011ee2 pushed by Hein-Htet-Aung08							
✓ Merge pull request #57 from Hein-Htet-Aung08/student-1-andy-staff-man...	main	Oct 6, 18:49 GMT+11	1m 23s	...			
Matthew - Performance and Professional Development #133: Commit 9aa3142 pushed by TLim05							
✓ Rag Eval updated.	student-2-hein-teaching-sub...	Oct 6, 18:37 GMT+11		...			

Individual Contributions

Features Updates/Release 1 Work

Student	Feature	Feature Updates and Release 1 Work
Andy Lam	Academic Staff Management	• Two MCP Tools - staff profile lookup and search by expertise. • Staff data added to shared RAG server. • Backend routes connecting to MCP and RAG servers. • AI assistant panel showing tool results, citations, confidence and raw structured JSON. • Updated docker-compose.yml to use host Ollama, MCP and RAG. • CI workflow with 52 pytests. • MCP and RAG validation modes for agentic loop.
Hein Htet Aung	Teaching, Subject and Classroom Allocation	• Added two MCP tools for allocation validation and classroom availability. • Added Student 2 data and joined allocation summaries to shared RAG. • Added MCP/RAG backend routes and UI tabs. • Integrated Staff and Workload services for staff validation/recommendations. • Added grounded RAG citations/confidence and retrieval benchmarks. • Extended CI for allocation rules, Docker/API smoke tests and disabled MCP/RAG validation.
Tristan Lim	Workload and Availability Management	• Four MCP tools: ◦ Staff count, ◦ Staff by status, ◦ Workload detail, ◦ Open alerts • Workload profiles and open alerts added to shared RAG server • Backend routes connecting to MCP and RAG servers (switchable off with MCP_Enabled and RAG_Enabled) • MCP Mode and RAG mode tabs showing tool results, citations and

		confidence • Updated docker-compose.yml to connect to the host MCP and RAG servers • CI workflow checking MCP and RAG are disabled during CI/CD
Nicholas Hatzidimitriou	Research and Grant Management	• Four read only tools ◦ Project count ◦ Projects by Department ◦ Project Grant summaries ◦ Research History • Project, grants and publications added to shared RAG server • Backend routes toggleable with MCP_ENABLED and RAG_ENABLED, returning 403 if off. • Benchmark queries for RAG Validation mode • MCP and RAG showing structured answer / JSON, citations and confidence.
Matthew Barnard	Performance and Professional Development	• Added two read-only shared MCP tools: staff development summary and training search by skill area. • Added RAG context for reviews, goals, programs, enrolments and recommendations, with source metadata and six retrieval benchmarks. • Added backend clients/routes and MCP Tools/RAG Questions tabs displaying results, citations, confidence and insufficient context. • Updated connection configuration and CI while retaining CRUD and AI-Mode.

Integration and Validation Activities

Student 1 (Andy Lam)

Integration

- Added two MCP tools on the shared server, being staff profile and search by expertise.
- Added staff management context to the shared RAG server which includes staff profiles, availability and summary.
- Wired backend to MCP and RAG to new routes and added AI assistant panel to frontend.
- Added MCP, RAG and Ollama connection settings to docker-compose.yml via host.docker.internal.

Validation

- The MCP was tested through the UI and API, showing correct profiles, or otherwise displaying error codes.
- The RAG returns grounded answers along with a confidence rating and citations, or otherwise shows “insufficient context”.
- Additionally both servers were validated in the terminal via the curl command.
- 52 pytest cases passed with MCP and RAG disabled.
- Runs end to end through docker-compose.yml with the host MCP, RAG and Ollama.
- The agentic loop runs correctly in MCP and RAG validation modes.

Student 2 (Hein Htet Aung)

Integration

- Connected Student 2 backend/UI to shared MCP and RAG services
- Registered two MCP tools with existing allocation APIs
- Added allocation summary chunks and exposed Student 2 context to shared RAG
- Integrated Student 1 staff/expertise and Student 3 workload/availability services
- Configured Docker services to use host Ollama, MCP and RAG

Validation

- UI/API testing for validation of MCP allocation and availability of classrooms
- Tested grounded RAG retrieval, answers, citations and confidence
- Added 7 RAG benchmarks for student 2
- Executed automated tests and smoke tests during runtime to validate allocation rules
- Student 2 CI passed with MCP/RAG disabled
- Shared MCP and RAG agentic validation reports passed

Student 3 (Tristan Lim)

Integration

- Added four read only MCP tools on the shared server: staff count, staff by status, workload detail and open alerts.
- Added workload context to the shared RAG server, being summary counts, each staff workload profile and each open alert.
- Wired the backend to MCP and RAG.
- Added MCP Mode and RAG mode tabs to frontend
- Connected to the host MCP and RAG servers in docker-compose.yml via host.docker.internal.

Validation

- MCP was tested through UI and terminal, returning the correct staff or a 403 error when MCP mode is off.
- RAG returns grounded answers with citations and a confidence rating or insufficient context for unrelated questions
- student-3-TristanLim.yml pass with MCP and RAG disabled
- Release 0 workload, CRUD and AI-Mode checks still pass in the same workflow

Student 4 (Nicholas Hatzidimitriou)

Integration

- Created and implemented MCP tools in the shared server: project count, projects by department, project grants summary and research history. Each of them are read only.
- Added research and grant context to shared RAG server. Implemented a loader that turns projects, grants and publications into searchable chunks and are tagged with source and domain.
- Wired the backend to MCP (through MCP client over streamable-http) and RAG. Both MCP and RAG have tabs in the frontend.
- Added MCP and RAG settings to the docker-compose.yml file using host.docker.internal

Validation

- MCP was tested in both the UI and terminal, ensuring correct results were returned for valid inputs.
- RAG returns grounded answers with citations and a confidence category. Alternatively insufficient context if there is no supporting evidence attached.
- Shared RAG and MCP endpoints were also validated + Agentic loop running correctly.
- student-4-NicholasHatzidimitriou.yml passes with MCP and RAG disabled.

Student 5 (Matthew Barnard)

Integration

- Registered shared MCP tools and the Student 5 RAG loader.
- Frontend requests pass through Nginx and the backend; tools/loaders read existing APIs.
- Integrated Compose connects to host services through host.docker.internal.

Validation

- All 50 tests passed: 28 backend, 12 database, six MCP and four RAG-context tests.
- JavaScript syntax passed.
- CI on main passed tests, three image builds and smoke checks with AI-Mode/MCP/RAG disabled.
- Stored retrieval results show recall@5 of 1.0 for all six Student 5 questions.

GitHub Commit Logs (Individual)

Student 1 (Andy Lam)

Commits on Oct 6, 2026	
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management Alamb-beyond committed 1 hour ago · ✓ 16 / 16	654da6c
Use host Ollama for all backends and remove ollama containers Alamb-beyond committed 1 hour ago	4d6c825
Commits on Oct 1, 2026	
Merge pull request #52 from Hein-Htet-Aung08/student-5-matthew-performance-and-professional-development Alamb-beyond authored 5 days ago · ✓ 7 / 7	Verified 826ac14
Tell RAG generator to list every match for who/which questions Alamb-beyond committed 5 days ago · ✓ 2 / 2	185d40a
Request 8 RAG chunks per Staff Management question Alamb-beyond committed 5 days ago	d3ee8f9
Merge pull request #47 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation Alamb-beyond authored 5 days ago · ✓ 4 / 4	Verified c5837b0

Commits on Sep 30, 2026

Raise Student 4 RAG answer timeout to 120s	8f3e623		
Default shared RAG server to llama3.1:8b	4cac4d4		
Add status rosters and shorten department rosters in Student 1 RAG context	c063958		
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management	a38c15c		
Merge pull request #43 from Hein-Htet-Aung08/student-3-tristan-workload-and-availability-management Verified	86793af		
Add Student 1 pytest suite and extend CI for Release 1	9769cb1		
Add AI Assistant panel for MCP tools and RAG answers to Staff frontend	f41b266		
Point Staff Management AI-Mode at host Ollama	61364c6		
Reword Student 1 roster and summary RAG chunks	3db80ae		
Set temperature 0 for RAG grounded answer generation	4610700		
Connect Staff Management backend to shared MCP and RAG servers	ccb9287		
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management	07b638a		
Merge pull request #41 from Hein-Htet-Aung08/student-3-tristan-workload-and-availability-management Verified	4630754		
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management	57565ca		
Merge pull request #42 from Hein-Htet-Aung08/student-4-nicholas-research-and-grant-management Verified	cd935dd		
Fix RAG hash embedding only using the first 32 dimensions	4543850		
Merge remote-tracking branch 'origin/student-4-nicholas-research-and-grant-management' into student-1-andy-staff-management	31fd310		
Add Student 1 RAG context loader and retrieval benchmarks	1685479		

Commits on Sep 29, 2026

Add Student 1 MCP tools for staff profile and expertise search	c893059		
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Student 2 (Hein Htet Aung)

Commits

student-2-...

Hein-Htet-...

All time

Commits on Oct 7, 2026

Merge pull request #60 from Hein-Htet-Aung08/main

Verified3cdd10a

Hein-Htet-Aung08 authored 2 minutes ago

Commits on Oct 6, 2026

Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation

Verified894108b

Hein-Htet-Aung08 authored 6 hours ago · 9 / 9

Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation

9011ee2

Hein-Htet-Aung08 committed 6 hours ago · 15 / 15

Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation

65dbb83

Hein-Htet-Aung08 committed 6 hours ago

Rag validation report updated.

eb576d1

Hein-Htet-Aung08 committed 6 hours ago

Rag Eval updated.

7ba0532

Hein-Htet-Aung08 committed 6 hours ago · 3 / 3

Merge pull request #56 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation

Verified01b54df

Hein-Htet-Aung08 authored 7 hours ago

New reports for mcp and rag validation is produced.

9f4545a

Hein-Htet-Aung08 committed 7 hours ago

Merge pull request #55 from Hein-Htet-Aung08/main

Verified6d0cc77

Hein-Htet-Aung08 authored 7 hours ago · 6 / 6

Commits on Oct 1, 2026

Merge pull request #49 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation

Verifiedb77051b

Hein-Htet-Aung08 authored 5 days ago · 4 / 4

UX improved.

8c81e14

Hein-Htet-Aung08 committed 5 days ago · 7 / 7

UX improved.

ff9bf15

Hein-Htet-Aung08 committed 5 days ago · 3 / 3

Frontend UI improvements.

e523136

Hein-Htet-Aung08 committed 5 days ago · 3 / 3

Frontend and RAG improved.

c02444a

Hein-Htet-Aung08 committed 5 days ago · 3 / 3

Frontend edits.

c505196

Hein-Htet-Aung08 committed 5 days ago · 3 / 3

Student 2 CI improved for Release 1.

7ee5b06

Hein-Htet-Aung08 committed last week · 7 / 7

RAG agentic loop finalized.

055c598

Hein-Htet-Aung08 committed last week

RAG validation report updated. Hein-Htet-Aung08 committed last week	3c0d2b3		
RAG added to agentic loop. Hein-Htet-Aung08 committed last week	42f124c		
Rag Eval is updated and retrieval metrics are derived. Hein-Htet-Aung08 committed last week	dca467f		
Rag Pipeline updated to have better retrieval and AI Prompt improved for better answer generation. Hein-Htet-Aung08 committed last week	385b27f		
Commits on Sep 30, 2026			
RAG tab is added for the frontend. Hein-Htet-Aung08 committed last week · 3 / 3	d0ae84b		
Docker Compose and Backend updated for RAG. Hein-Htet-Aung08 committed last week · 3 / 3	4c2589c		
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 committed last week · 3 / 3	dbeb5a9		
RAG server partially set up for Student 2 and one llama is installed for the RAG server. Hein-Htet-Aung08 committed last week · 3 / 3	92a0721		
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 committed last week · 3 / 3	672c0d0		
MCP server setup withing backend and frontend of personal webpage. Hein-Htet-Aung08 committed last week · 3 / 3	d2fb54c		
MCP server set up for student 2. Hein-Htet-Aung08 committed last week	7527003		
Commits on Sep 28, 2026			
Merge pull request #40 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 authored last week Verified	19ba40d		
Fill out all common code files and dissect shared files into different sectors for different students to fill out. Intentionally leave some files that need implementation first empty. Hein-Htet-Aung08 committed last week	1424041		
Merge pull request #39 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 authored last week Verified	f18c43f		
Fix Release 1 scaffold JSON placeholder. Hein-Htet-Aung08 committed last week	ca499ec		
Report sections and current possible plain files are added to the scaffold. Hein-Htet-Aung08 committed last week	22b83bb		
Scaffold updated for Release 1. Hein-Htet-Aung08 committed last week	91dd96c		
Scaffold updated for Release 0. Hein-Htet-Aung08 committed last week	96491f9		
Commits on Sep 28, 2026			
Merge branch 'main' of https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD into student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 committed last week	1f4e250		

Student 3 (Tristan Lim)

Commits on Oct 7, 2026	Merge pull request #59 from Hein-Htet-Aung08/student-1-andy-staff-management Verified ffee5d3
Commits on Oct 6, 2026	Merge pull request #57 from Hein-Htet-Aung08/student-1-andy-staff-management Verified 9aa3142
Commits on Oct 3, 2026	Merge pull request #53 from Hein-Htet-Aung08/student-4-nicholas-research-and-grant-management Verified 5346fe1
Commits on Oct 1, 2026	Merge pull request #48 from Hein-Htet-Aung08/student-3-tristan-workload-and-availability-management Verified 326538b Fixed raw Json output 68dadc5
Commits on Sep 30, 2026	Merge pull request #46 from Hein-Htet-Aung08/student-3-tristan-workload-and-availability-management Verified f4f7f13 Merge pull request #45 from Hein-Htet-Aung08/student-1-andy-staff-management Verified 73a4a8a Merge pull request #44 from Hein-Htet-Aung08/student-1-andy-staff-management Verified a7e6988 performed cleanup 09d0e4c MCP and RAG fixed to route through shared servers 122164d Merge branch 'main' into student-3-tristan-workload-and-availability-management Verified f9c0826 RAG prompts 675174c RAG frontend 070fd65 RAG backend added a9d5ec7 RAG server created b63fff9 MCP prompts 1f07d39 MCP frontend ad98afb
Commits on Sep 29, 2026	MCP backend 573132f
Commits on Sep 28, 2026	Created MCP server cda8cb1

Student 4 (Nicholas Hatzidimitriou)

Commits on Oct 3, 2026		
Merge pull request #54 from Hein-Htet-Aung08/student-4-nicholas-research-and-grant-management		
NicholasHatzzi authored 3 days ago · 22 / 22		
fixed up rag_mode route to display error when the box isn't ticked. Fixed up individual docker-compose.yml part to mirror group members. added in mcp and rag workflows that were forgotten		
NicholasHatzzi committed 3 days ago · 23 / 23		
Commits on Oct 2, 2026		
Merge remote-tracking branch 'origin/main' into student-4-nicholas-research-and-grant-management		
NicholasHatzzi committed 4 days ago		
fixed mcp validation mode to invoke all registered tools and write a report. Implemented RAG answer system prompt for full multi-domain corpus		
NicholasHatzzi committed 4 days ago · 7 / 7		
Commits on Sep 30, 2026		
Changed MCP server and RAG server implementation to be more in line with the assignment specification and how the group wants it to be		
NicholasHatzzi committed last week · 9 / 9		
Commits on Sep 29, 2026		
Fixed backend import and connected MCP and RAG tabs to their shared servers. Fixed display issue that was causing [object Object] to generate during AI generation		
NicholasHatzzi committed last week · 3 / 3		
Updated Rag pipeline and eval to include individual feature addition, the same as MCP. Retrieval metrics is added for RAG search quality. Tools-contract.md explains what each shared RAG function is...		
NicholasHatzzi committed last week		
Added individual part into the shared mcp-server		
NicholasHatzzi committed last week · 3 / 3		
Merge branch 'main' into student-4-nicholas-research-and-grant-management		
NicholasHatzzi committed last week		
removed old files and switched to new ones		
NicholasHatzzi committed last week		
removed old files and switched to new ones		
NicholasHatzzi committed last week		
Moved MCP and RAG servers to shared ai-services/. Created subfolders to hold the tools.py and loader.py files to add individual logic without editing the shared pipeline		
NicholasHatzzi committed last week · 3 / 3		
Commits on Sep 27, 2026		
Added new feature into frontend index page and removed old css file from release 0 that was used for testing		
NicholasHatzzi committed last week · 3 / 3		
Commits on Sep 25, 2026		
Fixed AI prompts due to hallucination and HTML		
NicholasHatzzi committed 2 weeks ago · 3 / 3		
Followed RAG Server schematic from labs. Implemented Rag Server and linked it to backend and frontend.		
NicholasHatzzi committed 2 weeks ago · 3 / 3		
Commits on Sep 24, 2026		
Added MCP server integration and connected MCP-mode tab to backend and frontend as per the labs		
NicholasHatzzi committed 2 weeks ago · 3 / 3		

Student 5 (Matthew Barnard)

Commits on Oct 7, 2026
Merge pull request #61 from Hein-Htet-Aung08/student-5-matthew-performance-and-professional-development MattBarnard4 authored 7 minutes ago · ✓ 3 / 3 Verified70b090e
Commits on Oct 1, 2026
Merge pull request #51 from Hein-Htet-Aung08/student-1-andy-staff-management MattBarnard4 authored last week · ✓ 4 / 4 Verifiedb521661
Commits on Sep 9, 2026
Merge pull request #38 from Hein-Htet-Aung08/student-5-matthew-performance-and-professional-development MattBarnard4 authored last month Verifiedc9365bf
Merge pull request #33 from Hein-Htet-Aung08/student-5-matthew-performance-and-professional-development MattBarnard4 authored last month · ✓ 3 / 3 Verified37643ee
Merge pull request #32 from Hein-Htet-Aung08/student-5-matthew-performance-and-professional-development MattBarnard4 authored last month · ✓ 3 / 3 Verifieda54c3f4
Commits on Sep 3, 2026

Commits on Oct 7, 2026
Use host Ollama in Student 5 configuration Mattonut committed 27 minutes ago · ✓ 6 / 6 7ab7295
Rename RAG.drawio.png to RAG_release1.drawio.png NicholasHatzl authored 14 hours ago Verified982b1ff
Rename MCP.drawio.png to MCP_release1.drawio.png NicholasHatzl authored 14 hours ago Verified4f731a7
mcp and rag architecture images NicholasHatzl authored 14 hours ago Verified79fd35a
Merge pull request #59 from Hein-Htet-Aung08/student-1-andy-staff-management TLim05 authored 15 hours ago Verifiedflee5d3
Commits on Oct 6, 2026
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management Alamb-beyonder committed 18 hours ago · ✓ 4 / 4 9060c06
Add Release 1 integrated architecture diagram Alamb-beyonder committed 18 hours ago 8a0d351
Merge pull request #58 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 authored 20 hours ago · ✓ 9 / 9 Verified894108b
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 committed 20 hours ago · ✓ 15 / 15 9011ee2
Merge pull request #57 from Hein-Htet-Aung08/student-1-andy-staff-management TLim05 authored 20 hours ago · ✓ 12 / 12 Verified9aa3142
Merge remote-tracking branch 'origin/main' into student-2-hein-teaching-subject-and-classroom-allocation Hein-Htet-Aung08 committed 20 hours ago 65dabb83
Rag validation report updated. Hein-Htet-Aung08 committed 20 hours ago eb576d1
Rag Eval updated. Hein-Htet-Aung08 committed 20 hours ago · ✓ 3 / 3 7ba0532

Hein-Htet-Aung08 committed 20 hours ago	eb5db83	<>
Rag validation report updated.	eb576d1	<>
Hein-Htet-Aung08 committed 20 hours ago		
Rag Eval updated.	7ba0532	<>
Hein-Htet-Aung08 committed 20 hours ago · ✓ 3 / 3		
Merge remote-tracking branch 'origin/main' into student-1-andy-staff-management	654da6c	<>
Alamb-beyonder committed 20 hours ago · ✓ 16 / 16		
Use host Ollama for all backends and remove ollama containers	4d6c825	<>
Alamb-beyonder committed 20 hours ago		
Merge pull request #56 from Hein-Htet-Aung08/student-2-hein-teaching-subject-and-classroom-allocation	01b54df	Verified <>
Hein-Htet-Aung08 authored yesterday		
New reports for mcp and rag validation is produced.	9f4545a	<>
Hein-Htet-Aung08 committed yesterday		
Merge pull request #55 from Hein-Htet-Aung08/main	6d0cc77	Verified <>
Hein-Htet-Aung08 authored yesterday · ✓ 6 / 6		
Commits on Oct 3, 2026		
Merge pull request #54 from Hein-Htet-Aung08/student-4-nicholas-research-and-grant-management	7844279	Verified <>
NicholasHatzl authored 4 days ago · ✓ 22 / 22		
fixed up rag_mode route to display error when the box isn't ticked. Fixed up individual docker-compose.yml part to mirror group members. added in mcp and rag workflows that were forgotten	3d63d1e	<>
NicholasHatzl committed 4 days ago · ✓ 23 / 23		
Merge pull request #53 from Hein-Htet-Aung08/student-4-nicholas-research-and-grant-management	5346fe1	Verified <>
TLim05 authored 5 days ago · ✓ 4 / 4		
Commits on Oct 2, 2026		
Merge remote-tracking branch 'origin/main' into student-4-nicholas-research-and-grant-management	239a16d	<>
NicholasHatzl committed 5 days ago		
fixed mcp validation mode to invoke all registered tools and write a report. Implemented RAG answer system prompt for full multi-domain corpus	55c912a	<>
NicholasHatzl committed 5 days ago · ✓ 7 / 7		

Contribution Logs

Student Number	Name	Contribution
25249309	Andy Lam	- Created and scaffolded Release 1 Report - Release 1 Architecture - Validation and Results - Individual Sections and Feature + MCP tools and RAG context loader - Agentic Loop Validation
25026511	Hein Htet Aung	- Created initial Release 1 MCP/RAG scaffold, prompts, tool contracts and report structure - Implemented shared MCP/RAG common infrastructure - Implemented Student 2 MCP tools, RAG context/chunking and frontend/backend integration - Integrated Staff/Workload services with Student 2 allocation and AI workflows - Implemented RAG validation in the shared agentic loop - Contributed MCP/RAG integration review and

		final shared validation evidence • Wrote Release 1 Requirements • Extended Student 2 CI, automated tests and runtime/API validation
25299672	Tristan Lim	• Project Overview and Release 1 Scope • Individual sections and Feature • Individual MCP tools and RAG context loader
25471398	Nicholas Hatzidimitriou	• Release 1 RAG and MCP architecture • Individual Sections and Feature • Individual MCP tool and RAG loader • Contributed to RAG/MCP agentic loop integration • Implemented individual RAG context / chunking • RAG/MCP frontend and backend implementation
25563457	Matthew Barnard	• Implemented Student 5 MCP tools, RAG context and benchmarks, frontend/backend integration, configuration, tests and CI. • Prepared Non-Functional Requirements

Showcase video URL

<https://www.youtube.com/watch?v=4aZCjALL2q0>

GitHub Repository Link

<https://github.com/Hein-Htet-Aung08/faculty-management-system-ASD>

Known Issues and Limitations

Academic Staff System (Andy Lam)

- MCP, RAG server and Ollama are required to be manually started in the host.
- Agentic loop is run from command line and its MCP and RAG validation modes require their respective servers and Ollama to be running.
- Currently there are only two read-only tools which are staff profile and search by expertise. These tools are chosen through their specific button and not an LLM's driven tool selection.
- There is an absence of authentication on the local MCP server.
- The index isn't updated automatically and requires the user to press "Refresh Knowledge Base", rebuilding the shared index.
- The retrieval currently only operates on words and not semantic meanings so synonyms aren't properly considered.

- The list answers depends on the top-k (8 chunks), which can leave entries out.
- A wrong answer can have a high confidence as the rating measures source quality.

Teaching, Subject and Classroom Allocation (Hein Htet Aung)

- MCP, RAG and Ollama must be running locally on the host, otherwise, Student 2 MCP/RAG functions are not available.
- RAG data is not updated automatically after the change in the database and needs to be refreshed in the knowledge-base.
- Retrieval prioritizes recall over precision, so irrelevant chunks might appear and some expected evidence might be missed at top-k.
- RAG/LLM responses can be slow, with a 120 second timeout on answer requests.
- Student 2 MCP tools only validate allocations / check availability, they do not create allocations or book classrooms.
- CI runs with MCP/RAG turned off, so live shared-service correctness is validated separately via UI/API, terminal and agentic-loop testing.

Workload and Availability Management (Tristan Lim)

- MCP and RAG are disabled in CI, the workflow only proves they are switched off, not that the results are correct.
- The four tools are read only and run from fixed buttons in the MCP tab rather than being chosen by an LLM.
- New or changed workload data only reaches RAG after the knowledge base is refreshed.
- LLama can take long enough on a first question to hit the 120 second request timeout.

Research and Grant Management (Nicholas Hatzidimitriou)

- The LLM can misattribute details even if the retrieval is correct: if I ask for three staff members associated with a grant, RAG will sometimes only retrieve one.
- Precision is inconsistent and varies a lot compared to how it should be. The current range is between 0.2 and 0.8 depending on the question being asked, but the retrieval averages out to 0.9-1, often because of irrelevant chunks.
- Both RAG and MCP require manual startups.
- The Agentic Loop needs both RAG and MCP running or else it fails first call
- Llama takes a minute or longer to load on first use for RAG Server.

Performance and Professional Development (Matthew Barnard)

- MCP, RAG and Ollama require host startup.
- Changed records require a corpus refresh.
- Top-five retrieval can omit evidence; citations list retrieved sources and confidence is heuristic.
- Model loading may exceed request timeouts.
- CI uses mocks and disabled modes, so live demonstrations remain necessary.